Existence and Enumeration of Symmetric Cube Colorings via a Linear Algebra System

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Abstract

We find a specific solution for the symmetric case of a problem of coloring the faces of an $n \times n \times n$ cube with n distinct colors. Our solution allows this cube to be rearranged into n distinct configurations such that the entire surface is colored with each distinct color. We create several rigorous abstractions from the original problem, culminating in a simple linear algebra system which describes all symmetric solutions for a given n and provide an implementation of Google's CP-SAT solver to enumerate these solutions efficiently. Additionally, we show the existence of a solution for arbitrarily large n.

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1 Introduction

Consider a cube of side-length n, which comprises n^3 individual unit cubes. The problem is to color every face on these unit cubes with one of n distinct colors in such a way that the overall cube may be rearranged, via rotation or repositioning of the unit cubes, into n distinct configurations such that an arrangement exists where the cube's entire surface area is colored entirely by each of the n distinct colors.

This paper has two main motivating questions: is it possible to arrange n colors on any arbitrarily large $n \times n \times n$ cube so that these constraints are satisfied? And if so, how many solutions exist for a given n and may we generate them?

2 Lemmas and Definitions

We consider a cube with side-length n, composed of n^3 individual unit cubes. The total surface area of the cube comprises $6n^2$ faces. Any face on the cube may be colored by a single color c_i and no face may be occupied by more than one color.

The abstractions and proofs presented in this paper are valid for all cube sizes $n \geq 2$. The trivial case for $n = 1$ is excluded as the arrangement of the faces on its surface area is not consistent with the general case for $n \geq 2$.

Definition 2.1. A valid solution for n is a coloring of the faces of n^3 unit cubes with n distinct colors such that, for each of the n colors, the set of unit cubes can be rearranged, via repositioning or rotating them, to form a cube with side-length n where the surface area of the cube is colored entirely by that color.
Assumption 2.2. In order to color the entire surface area of a cube, a given color must occupy at least $6n^2$ faces.
Assumption 2.3. The number of colored faces on a cube may not exceed $6n^3$, or the total number of faces on that cube.
Lemma 2.4. *The total number of faces occupied by each color is exactly* $6n^2$

Proof. Let n be the side-length of the cube that is being colored and let C be the set of all n distinct colors.

Let $F(c)$ give the number of faces that the color $c \in C$ occupies. By Assumption 2.2, $F(c) \geq 6n^2 \quad \forall c \in C$. There are n distinct colors, so the lower bound of $\sum_{c \in C} F(c) = 6n^2 \cdot n = 6n^3$.

It can then be shown that $F(c) \leq 6n^2$ using a proof by contradiction. Assume that there exists some $c_i \in C$ such that $F(c_i) = 6n^2 + k \quad (k > 0)$. Let $C' = C - \{c_i\}$. Since $F(c) \geq 6n^2$, the minimum value of $\sum_{c \in C'} F(c)$ is

$6n^2 \cdot |C'| = 6n^2(n-1)$. Because $C' \cup c_i = C$, the previous summation can then be split into two distinct summations:

$$\begin{aligned}
\sum_{c \in C} F(c) &= \sum_{c \in C'} F(c) + \sum_{c \in \{c_i\}} F(c) \\
&= \sum_{c \in C'} F(c) + F(c_i) \\
&= \sum_{c \in C'} F(c) + 6n^2 + k \\
&= 6n^2(n-1) + 6n^2 + k \\
&= 6n^3 + k
\end{aligned}$$

$6n^3 + k > 6n^3$, which violates Assumption 2.3. The proposition that $F(c) > 6n^2$ leads to a contradiction, which implies that $F(c) \leq 6n^2$ must be true.

Therefore, $\forall c \in C$, $6n^2 \leq F(c) \leq 6n^2$ and $F(c) = 6n^2$ □

Corollary 2.5. *All faces on the cube must be colored in order to form a solution for n colors in a cube with side-length n.*

Proof. By Lemma 2.4, each color must occupy $6n^2$ faces. There are n distinct colors, so the total number of faces occupied is $6n^2 \cdot n = 6n^3$. This is equivalent to the total number of faces on the entire cube, so therefore all faces of the cube must be colored with one of the n distinct colors for the resulting configuration to form a valid solution. □

Lemma 2.6. *Every face of a given color must appear on the cube's exterior simultaneously to form a valid solution*

Proof. Let c_i be a given color and n be the side-length of the cube for which a solution is constructed.

By Lemma 2.4, the number of faces with the color c_i is $6n^2$. Additionally, the surface area of a cube comprises $6n^2$ faces. This means that, for c_i to form part of a valid solution, per Definition 2.1, every colored face must appear on the cube's exterior because the number of available faces equals the number of required faces. □

Lemma 2.7. *A cube with r faces of color c_i must be arranged such that r faces are exposed in a valid solution for a given color*

Proof. Let c_i be a given color and n be the side-length of the cube for which a solution is constructed. Let f be the number of faces of a given unit cube that are exposed on the cube's surface area.

Consider the case where the cube is arranged such that $r - k \quad (k > 0)$ of its faces are exposed. In this case, k faces of color c_i are interior faces which do not comprise part of the cube's exterior. By Lemma 2.6, this cube therefore may not form part of a valid solution. Thus, $f \geq r$.

Additionally, consider the case where the cube is arranged such that $r + k \quad (k > 0)$ of its faces are exposed. In this case, k of the faces comprising the cube's surface area will not be of color c_i. By Definition 2.1, this solution is therefore not valid because it violates the condition that the entire surface area is comprised of faces with color c_i. Therefore, $f \leq r$.

Finally, $r \leq f \leq r$ and so $f = r$. □

Lemma 2.8. *Each unit cube may have no more than 3 faces of the same color in a valid solution*

Proof. Let c_i be a given color.

Consider a unit cube that has k faces of the color c_i where $k > 3$. Because no more than 3 faces of a unit cube comprise part of the cube's surface, $3 - k$ of the faces on this cube will always be internal faces which cannot comprise part of the cube's surface. This violates Lemma 2.6, thus it impossible for c_i to form part of a valid solution, therefore invalidating the entire solution. This shows that each unit cube in a valid solution must have no more than 3 faces of the same color. □

Lemma 2.9. *The set of all faces on a unit cube which comprise part of the surface area of the overall cube must be mutually adjacent*

Proof. Let C be any unit cube which comprises part of the surface area of the larger cube. Let n be the side-length of the larger cube. Let F be the set of all faces on C which comprise part of the overall cube's surface area. By Lemma 2.8, $|F| \in \{1, 2, 3\}$.

In the case where $|F| = 1$, this lemma's statement is vacuously true because, when the set contains only 1 face, there are no other faces in the set for which this face could not be mutually adjacent. Thus, the statement holds for this trivial case.

We now proceed to prove the statement for the non-trivial cases of $|F| = 2$ and $|F| = 3$.

A cube with side-length n is composed of a set of unit cubes. The position of each unit cube may be represented by a co-ordinate vector $V = (x, y, z)$ which represents the distance of this cube's center from the center of the larger cube along the x, y and z axes. We assume the larger cube's center to be at the origin $(0, 0, 0)$.

The total width of the cube is n, so the maximum absolute distance from the cube's center to any of its faces is $\frac{n}{2}$. The faces with the highest distance are those which comprise the cube's surface area, because this represents the edge of the cube. The unit cubes which comprise the cube's surface area have centers at co-ordinates such that the absolute distance to any of their faces is $\frac{1}{2}$. For the unit cubes to be fully contained within the larger cube, these centers V_i must not be outside the cube's boundaries. Thus, they must satisfy $|V_i| \leq \frac{n}{2} - \frac{1}{2} = \frac{n-1}{2}$.

Therefore, the co-ordinates of the centers of unit cubes must not exceed a distance of $\frac{n-1}{2}$ along any axis. Additionally, in order to comprise of faces which form part of the cube's surface area, a given unit cube must have at least one co-ordinate equal to $\frac{n-1}{2}$ because otherwise it would be impossible for its faces to reach the overall cube's boundary.

Consider each direction in which a cube may border C to be a translation in space from the center of C to the center of some other cube as a vector $T = (x, y, z)$ where x, y and z are the x, y and z co-ordinates of the translation and $|x + y + z| = 1$. More specifically, the cubes directly bordering C may be described by the translations $\pm(1, 0, 0)$, $\pm(0, 1, 0)$ and $\pm(0, 0, 1)$. A given translation T specifies a unit cube which borders C if and only if it does not produce co-ordinates which violate $|T_i| \leq \frac{n-1}{2}$, as defined above (as this would not be the center of a valid unit cube within the boundaries of the overall cube.)

The statement that all faces in F are mutually adjacent is equivalent to the statement that, for any subset $S \subseteq F$ where $|S| = 2$, the two faces contained in S are adjacent. To prove this fact, we first assume the opposite: there exists a subset $S \subseteq F$ with $|S| = 2$ where the two faces contained in S are not adjacent. Two faces on a unit cube which are not adjacent must therefore be opposite, so the faces in S are opposite faces.

Both faces in S do not border the face of another cube because, if this were the case, they would be interior faces which would not comprise part of the overall cube's surface area. This implies that there exists no cubes directly bordering C in two opposing directions. These two opposing directions are then the positive and negative

translations in one axis (x, y or z). Therefore, the cubes bordering C must only border along at most two distinct axes.

Claim 2.10. *In a cube with side-length n, any unit cubes comprising the surface area with 2 or 3 faces exposed are bordered along 3 distinct axes.*

Proof. Let $m = \frac{n-1}{2}$. This is the upper bound for the co-ordinates of a cube's center along any axis.

First, we prove the case for a cube with 3 faces exposed. A unit cube with 3 faces exposed is a corner piece of a larger cube. Corner pieces are placed at the locations which are furthest from the cube's center, so their co-ordinates are (c_1m, c_2m, c_3m) where $c_n \in \{1, -1\}$. It then follows that the three translations $-c_1(1, 0, 0)$, $-c_2(0, 1, 0)$ and $-c_3(0, 0, 1)$ produce valid unit cubes, because each reduces to a translation of c_n in the respsective co-ordinate for that coefficient. That is, $c_nm - c_n = c_n(m - 1)$ which satisifes $|c_n(m - 1)| \leq m$ because $|c_n| = 1$. These three translations all translate along a distinct axis, so cubes with 3 faces exposed are bordered along 3 distinct axes.

Next, we prove the case for a cube with 2 faces exposed. This is an edge piece, which lies on an edge of the cube whose distance is as far as possible (that is, m) for two of its three co-ordinates. Therefore, and edge piece will have two co-ordinates with absolute value m and one co-ordinate with absolute value $\leq m - 1$, because if the absolute value of the third co-ordinate were also m, this cube would instead be a corner piece and have 3 faces exposed.

We can represent the co-ordinates of an edge piece with $V = (c_1v_1, c_2v_2, c_3v_3)$ where $\{v_1, v_2, v_3\} = \{m, m, k\} \quad (k \leq m - 1)$. Regardless of which $v_n = k$, we may apply the same logic as the case for the cube with 3 exposed faces: the translations $-c_1(1, 0, 0)$, $-c_2(0, 1, 0)$ and $-c_3(0, 0, 1)$ produce $c_nv_n - c_n = c_n(v_n - 1)$. The maximum value of $v_n - 1$ is $m - 1$, therefore, for any value of v_n, $|c_n(v_n - 1)| \leq m$. The three translations applied all translate across distinct axes so a cube with 2 faces exposed is bordered along 3 distinct axes.

One should note that, in reality, there exist four cubes which border a cube with 2 faces exposed: the final translation is the translation in the opposite direction for the dimension with $v_n = k$. However, a rigorous statement of this translation is not required for a valid form of this proof.

Thus, in the case where a cube has 2 or 3 faces exposed, it is bordered along 3 distinct axes. □

The validity of this claim then produces a direct contradiction to the implication that the cubes bordering C must only border it along at most two distinct axes. Therefore, the assumption that there exists a set S with two non-adjacent faces implies a contradiction and is false, and thus it must be the case that all faces in F are mutually adjacent. □

Lemma 2.11. *All faces of a given color must be adjacent to all other faces of the same color on a single unit cube in a valid solution*

Proof. Let c_i be a given color and n be the side-length of the cube for which a solution is constructed.

Consider a unit cube with a subset of its faces F, each of which have the color c_i. In order to prove the original statement, we first assume the opposite: that at least one face in F is not adjacent to all others. That is, the set of faces in F are not mutually adjacent.

By Lemma 2.9, this set of faces would therefore not all be able to comprise part of the cube's surface area simultaneously because they are not all mutually adjacent. This implies that at least one face in F is not exposed as part of the overall cube's surface area. This violates Lemma 2.6, because all faces with the color c_i, including those in F, must appear on the cube's surface area simultaneously to form a valid solution. Therefore, this arrangement of faces would prevent a valid solution from being formed.

Thus, in order to form part of a valid solution, all faces of color c_i on a given unit cube must be adjacent to all other faces of color c_i - that is, they must be mutually adjacent. $\square$

3 Single-color Solutions: b-constraints

Notation. A region on a cube is a set of mutually adjacent faces of the same color. A cube contains a region of size k with color c_i if and only if the cube contains exactly k faces of color c_i. A region is a single entity and the cube is not considered to simultaneously contain smaller subdivisions of the same region.

Given the defined lemmas above, we now show that a final condition on the solution space of unit cubes that can form part of a valid solution, a b-constraint, completes a set of conditions which are both necessary and sufficient so that they describe all possible valid solutions and only possible valid solutions for a single color.

A b-constraint, fundamentally, denotes the number of cubes of each region size that the surfacea area of a cube with side-length n comprises of: 8 cubes where 3 faces are exposed (corners); $12(n-2)$ cubes where 2 faces are exposed (edges); and $6(n-2)^2$ cubes where 1 face is exposed (center pieces). We then let b be a b-constraint where b_r gives the number of cubes with region size r:

$$b = \begin{pmatrix} 6(n-2)^2 \\ 12(n-2) \\ 8 \end{pmatrix}$$

Because the faces of all unit cubes on the exterior of the larger cube must be of color c_i for a valid solution to exist, a b-constraint can be used to constrain the amount of each region size that the set of all cubes with color c_i must contain. We now prove that this constraint is necessary for a valid solution to exist.

3.1 Necessity

Theorem 3.1. *For a given color c_i and region size $r \in \{1, 2, 3\}$, there must be exactly b_r unit cubes with regions of size r that are of color c_i*

Proof. Let r be a region size with $r \in \{1, 2, 3\}$. Let C_r be the set of all unit cubes with r faces of color c_i.

To find the lower bound for $|C_r|$, we use a proof by contradiction: suppose $|C_r| = b_r - k \quad (k > 0)$. Because the surface area of the cube comprises b_r cubes with r faces exposed, when the cube is rearranged to show only c_i on the surface of the cube, there will then be k unit cubes with r faces exposed that do not have r faces of color c_i. This violates Lemma 2.7, and therefore a valid solution will not exist in this case.

Therefore, $|C_r| \geq b_r \quad (\forall r \in \{1, 2, 3\})$.

To find the upper bound for $|C_r|$, we first compute the number of faces with color c_i using the summation $S = \sum_{x \in \{1,2,3\}} x \cdot |C_x|$. This holds because every cube in the set C_x contains x faces of the color c_i and all cubes with one or more faces of color c_i are contained in one of the sets C_x. The value of this summation can then be found

when $|C_x| = b_x$ for all $x \in \{1, 2, 3\}$:

$$\begin{aligned} 3 \cdot 8 + 2 \cdot 12(n-2) + 1 \cdot 6(n-2)^2 &= 24 + 24(n-2) + 6(n-2)^2 \\ &= 24(n-1) + 6(n-2)^2 \\ &= 24n - 24 + 6n^2 - 24n + 24 \\ &= 6n^2 \end{aligned}$$

We then consider the case where $|C_r| = b_r + k \quad (k > 0)$. Let R be the set $\{1, 2, 3\} - \{r\}$

$$\begin{aligned} \sum_{x \in \{1,2,3\}} x \cdot |C_x| &= \sum_{x \in R} x \cdot |C_x| + \sum_{x \in \{r\}} x \cdot |C_x| \\ &= \sum_{x \in R} x \cdot |C_x| + r \cdot b_r + r \cdot k \end{aligned}$$

The equation $\sum_{x \in R} x \cdot |C_x| + r \cdot b_r$ is equivalent to the value of S where $|C_r| = b_r$ for all r. Therefore, the above equation may be rewritten:

$$S + r \cdot k$$

The minimum value of S in this case is $6n^2$ by the previously established lower bound, so therefore

$$S + r \cdot k \geq 6n^2 + r \cdot k$$

Both r and k are nonzero and positive, thus the value of this summation is greater than $6n^2$. This contradicts Lemma 2.4, thus this set of faces could not form part of valid solution. This shows that, in order to form a valid solution, $|C_r| \leq b_r \quad (\forall r \in \{1, 2, 3\})$.

Finally, $b_r \leq |C_r| \leq b_r$ hence $|C_r| = b_r$. □

3.2 Sufficiency

In this section we show that any solution which satisfies a b-constraint and does not violate any previous Lemmas will always from a valid solution. This is an important result which will form the basis for the construction of solutions for n colors.

Theorem 3.2. *A set of unit cubes adhering to a b-constraint for a given cube size and color c_i is sufficient to form a valid solution for that cube size and for the color c_i.*

Proof. Let n be the side-length of the cube for which a solution is constructed. Let c_i be the color for which the solution is constructed. Let b be the b-constraint for the cube with side-length n.

Let $G = (V, E)$ be an undirected bipartite graph which denotes the positioning of unit cubes on the surface area of a cube with side-length n. $V = P \cup C$ where P is the set of positions on the surface area of the cube and C is the set of cubes with at least one face of color c_i (i.e., the set of cubes which may fill the positions in P when a solution is contructed for the color c_i). An edge $e \in E$ is created between a node $p \in P$ and a node $c \in C$ if and only if the cube c is placed at the position p. Only one cube may be assigned to any individual position.

The set P may be expressed as $P_3 \cup P_2 \cup P_1$ where P_k is the set of positions on the cube's surface area with k faces exposed. Then, $|P_k| = b_k$ where b is a b-constraint because b_k denotes the number of unit cubes on a larger cube with k faces exposed and this set of unit cubes adheres to a b-constraint.

Similarly, C may be expressed as $C_3 \cup C_2 \cup C_1$ where C_k is the set of cubes with k faces of color c_i. By Lemma 2.8, no cube mas this solution adheres to a b-constraint.

We then create 3 distinct subgraphs from the graph G, G_1, G_2 and G_3. $G_k = (V_k, E_k)$ where $V_k = P_k \cup C_k$ and E_k is the set of all edges in the graph G that are induced by the vertex set V_k (that is, the set of all edges which connect any two nodes in V_k). A cube may not be assigned to a position which is outside of its respective subgraph because this cube would then be arranged such that the number of faces it has of color c_i (k) would not match the number of faces exposed in its position, therefore violating Lemma 2.7.

Claim 3.3. *Given a subset G_k, any cube $c \in C_k$ may be rotated in any position $p \in P_k$ such that k faces of color c_i are exposed on c.*

Proof. Let f be the number of faces of color c_i on c (it is additionally the value of k in G_k.)

By Lemma 2.11, the set of faces on a c with color c_i must be mutally adjacent. Additionally, a cube c_p which exposes k faces at position p must also have all faces of color c_i mutually adjacent, per Lemma 2.9. If both c and c_p are arranged in such a way that satisfies these constraints, then the geometric constraints of faces with color c_i will be identical. Therefore, the relative positions of all others faces to these colored faces must also be identical. Therfore, the cubes are intrinsically geometrically identical and a rotation must exist between them. □

Given a subgraph G_k, each individual cube $c \in C_k$ may be assigned a unique position $p \in P_k$ because $|C_k| = |P_k|$. Thus, per Claim 3.3, c may be rotated so that all k faces on p are of color c_i. It follows that every face which resides on a position in P_k will then be of color c_i.

The set of positions which comprise all subgraphs ($P_3 \cup P_2 \cup P_1$) is equivalent to the set of all positions on the cube's surface area (P). Because every face on the cube's surface area resides in some position in $p \in P$, it then follows that every face on the cube's surface area will be of color c_i. By Definition 2.1, this implies a valid solution has been constructed for this color. □

4 n-color Solutions: UC vectors

In the previous section, we have shown that if the set of all unit cubes with at least one face of a given color satisfies its respective b-constraint, then this condition is sufficient to show that they can be arranged to form a valid solution for that color. However, this addresses solutions for only a single color at a time. The question remains whether there exists an arrangement of colored regions on all n^3 unit cubes so that, for all n distinct colors, the set of cubes with faces of that color satisfy their respective b-constraint (that is, whether a configuration of unit cubes exists so that there is a valid solution for all n colors using the same set of cubes). This is the issue that will be adressed in this section.

In order to generalise the solution to all n colors, we introduce an abstracted representation of any given unit cube, a unit cube vector (hereafter UC vector). Let $u \in \mathbb{N}^3$ be a vector of the form (r_1, r_2, r_3) where r_k represents the number of regions of size k on the unit cube. These configurations must color exactly six faces in total and must not exceed the total number of assignable colors. Thus, the constraints on a UC vector are shown below.

$$u = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} \qquad \text{with } 3r_3 + 2r_2 + r_1 = 6 \text{ and } r_3 + r_2 + r_1 \leq \min(n, 6)$$

Because there are n colors on a cube and each color requires a b-constraint of b, it follows that the constraint of 3-, 2- and 1-sized regions for a solution for all n colors would be $n \cdot b$. This is referred to hereafter as a global b-constraint, denoted by B. This section will then show that a set of n^3 unit cube vectors summing to a global b-constraint is a necessary and sufficient condition for a valid solution to exist in the symmetric case.

4.1 Necessity

Theorem 4.1. *Given a set of unit cubes forming a valid solution represented by a set of UC vectors U, the sum of U must equal B.*

Proof. Let n be the side length of the cube for which a solution is constructed. Let U be a set of unit cube vectors with $|U| = n^3$, per Corollary 2.5. Let B be the global b-constraint for the cube with side-length n. Let b be the b-constraint for the cube with side-length n.

Let $T = \sum_{u \in U} u$. Then, by definition, T_r represents the total number of regions in the set of unit cubes represented by U which are of size r because u_r for a UC vector $u \in U$ represents the number of regions of size r on the unit cube represented by u.

For each region size $r \in \{1, 2, 3\}$, per Theorem 3.1, each color must have b_r regions of this size. There are n distinct colors which occupy distinct regions, so therefore the total number of regions with size r is $n \cdot b_r$. This is equivalent to the value of $(n \cdot b)_r = B_r$. Thus, the total number of regions of size $r \in \{1, 2, 3\}$ must be equal to B_r over the entire cube. More specifically, $T_r = B_r \quad \forall r \in \{1, 2, 3\}$. Since both $|T| = |B| = 3$, it follows that $T = B$. This shows that the sum of U must equal B if the set of unit cubes represented by U forms a valid solution. □

4.2 Sufficiency

As the necessity of a global b-constraint has been established, we must now show that it is also a sufficient condition to form a valid solution for all colors. The author has not found a proof for the general case of any arbitrary set of UC vectors U, but a proof for the symmetric case of U, where the number of each distinct UC vector is a multiple of n, is proven below. In Section 5.3, we show that there exists at least one symmetric case for any $n \geq 6$, and thus a solution for arbitrarily large n still exists.

First, we provide a necessary intermediate proof for the number of regions on a unit cube.
Lemma 4.2. *The number of distinct regions on a given unit cube may not exceed* $\min(n, 6)$.

Proof. Let F be the set of all faces on a given unit cube with $|F| = 6$. Let C be the set of all distinct colors which a face in F may be colored with, where $|C| = n$. Let S be the set of all regions in F: that is, subsets of F containing a set of mutually adjacent faces of the same color.

The elements of S are pairwise disjoint because there exists only one region per distinct color, by Lemma 2.11. Thus, it follows that the total number of subsets may not exceed the number of faces on a cube: $|S| \leq 6$ as each face may not be colored with more than one color.

Additionally, as there exists at most one subset for each distinct color, $|S| \leq n$. The final condition is then:

$$\begin{aligned} &|S| \leq 6 \wedge |S| \leq n \\ \implies &|S| \leq \min(n, 6) \end{aligned}$$

□

Theorem 4.3. *If a set of unit cube vectors U satisfies the conditions that their sum is equal to B and that the number of each unique vector type in U is a multiple of n, then there exists a valid solution for the set of unit cubes represented by U.*

Proof. Let n be the side length of the cube for which a solution is constructed. Let U be a set of unit cube vectors with $|U| = n^3$, per Corollary 2.5. Let B be the global b-constraint for the cube with side-length n. Let b be the b-constraint for the cube with side-length n. Let C be the set of all n distinct colors.

By Theorem 3.2, if there exist b_k regions with color c_i of size k with $k \in \{1, 2, 3\}$ for all n colors c_i, a valid solution will exist for each of these colors and thus a valid solution will exist for the entire cube. The problem then reduces to showing that these required regions can validly be constructed on n^3 individual unit cubes.

If every $u \in U$ satisfies the constraints defined at the beginning of this section, it will be possible to place the quantities of regions described by u on a single unit cube. Coloring these regions is more difficult: each unit cube may have no more than one region of a given color because otherwise, when a region of that given color is exposed to the surface area, Lemma 2.7 would be violated as there would be other faces of the same color belonging to separate regions which would not be exposed. Thus, we must color every unit cube such that no two regions on the same unit cube have faces of the same color. We now show that this is possible for any set of unit cubes which follow our constraints.

Let R denote the set of all regions on the faces of the set of unit cubes which are represented by U. Let R_k give the set of regions with size k in R. $|R_k| = B_k$ because $\sum U_k = B_k$.

We then reduce the initial problem to a graph coloring problem with cardinality constraints. Let $G = (V, E)$ be an undirected graph representing the coloring of all regions in R. There exists an individual node $v_r \in V$ for each distinct region $r \in R_k$. Each individual unit cube represented by $u \in U$ forms a clique of size $\sum u_i$ consisting of all regions nodes v_r on the unit cube u. The constraint that two regions on the same cube may not share a color is enforced by the fact that every node in a clique connects to every other node in that clique.

Let $R_k(c_i)$ give the number of regions with size k whose faces are of color c_i. Then, we impose additional cardinality constraints on the graph to ensure that it represents a valid solution. Specifically, by Theorem 3.1,

$$\forall c_i \in C \ \forall k \in \{1, 2, 3\} \quad R_k(c_i) = b_k$$

If a graph is properly colorable and satisfies this condition, then it will represent a valid solution to the cube coloring problem.

Let T represent the set of unique UC vector types in U. Let $w(T_i)$ equal the number of UC vectors $u \in U$ with $u = T_i$. By definition, $n \mid w(T_i) \quad \forall T_i \in T$ and so $w(T_i) = mn \quad (m \in \mathbb{N})$

Let G_i be a subgraph of G that represents all the nodes represented by the UC vector T_i: that is, G_i is the subgraph induced by the vertex set $V_i = \{v_r \in V \mid r = T_i\}$.

G_i contains $w(T_i)$ distinct cliques. As $w(T_i) = mn$, we may then partition G_i into m distinct subgraphs $G_{i,m}$ containing n cliques each.

Let $C_{i,m}$ denote the set of cliques comprising the subgraph $G_{i,m}$. For each clique $c_j \in C_{i,m}$, let V be the set of vertices comprising it. Per Lemma 4.2, $|V| \leq min(n, 6)$, thus the values of the set V are derived by the indexes V_k with $0 \leq k < \sum T_i$ with $\sum T_i \leq \min(n, 6)$. Additionally, we number the vertices in V such that the size of the region represented by V_k is constant for all $c_j \in C_{i,m}$.

Then, the color of vertex V_k in clique c_j is:

$$C_{j+k \mod n}$$

We then show that this coloring pattern is sufficient to show that a valid coloring exists.

Claim 4.4. *This coloring ensures the graph is colored properly*

Proof. We first assume the opposite: that for some two distinct nodes V_{k_1} and V_{k_2} in the same clique have the same color. As their color is $C_{i+j \mod n}$, this implies that :

$$j + k_1 = j + k_2 \mod n$$
$$k_1 = k_2 \mod n$$

As $k < \min(n, 6)$, it follows that $k < n$. The only values of k_1 and k_2 that would satisfy the above relation is then the case where $k_1 = k_2$. However, this contradicts our initial assumption that V_{k_1} and V_{k_2} are distinct nodes. Thus, it must be that no nodes in the same clique share the same color, and thus the graph is properly colorable as all cliques are disconnected components. □

Claim 4.5. *This coloring satisfies* $R_k(c_i) = b_k \quad \forall c_i \in C \ \forall k \in \{1, 2, 3\}$.

Proof. Consider a subgraph $G_{i,m}$, which by construction contains n cliques c_j $(1 \leq j \leq n)$ constructed via the UC vector T_i. By definition, there are $(T_i)_k$ nodes in each clique representing a region with size k. As a node V_l represents a region of constant size for all cliques $c_j \in G_{i,m}$, it follows that there must be $(T_i)_k$ values of l such that the node V_l always represents a region with size k across all cliques $c_j \in G_{i,m}$.

Let J be the set of all clique indexes j where V_l is a member of c_j. $G_{i,m}$ is partitioned into n cliques, so by construction V_l is a member of all of them and their indexes are $J = \{1, 2, ..., n\}$. Thus, the coloring of V_l in all cliques is given by the complete residue system modulo n $\{j+l \mod n \mid j \in J\}$ where l is constant, which generates all values 1 through n. This shows that the node V_l is colored by all colors in C exactly once as $|C| = n$.

Thus, given a subgraph $G_{i,m}$ and any color $c_i \in C$, there are $(T_i)_k$ nodes that represent a region of size k colored with c_i. G_i contains m subgraphs of this kind, so this value is $m \cdot (T_i)_k$ for the entire subgraph G_i. This is equivalent to the value of $R_k(c_i)$ for the entire subgraph. Finally, we prove that this statement is equivalent to the original condition of this claim.

Because $w(T_i)$ represents the number of time T_i appears in U and $\sum U = B$, it follows that:

$$\sum_{T_i \in T} w(T_i) \cdot T_i = B$$

The summation can be split into the individual entries of T_i, thus:

$$\forall k \in \{1, 2, 3\} \sum_{T_i \in T} w(T_i) \cdot (T_i)_k = B_k$$

By definition, $w(T_i) = n \cdot m$ so we may rewrite the above equation and divide by n:

$$\forall k \in \{1, 2, 3\} \sum_{T_i \in T} nm \cdot (T_i)_k = B_k$$
$$\forall k \in \{1, 2, 3\} \sum_{T_i \in T} m \cdot (T_i)_k = b_k$$

We have shown above that this summation term is equivalent to $R_k(c_i)$, thus:

$$\forall k \in \{1,2,3\} \sum_{T_i \in T} m \cdot (T_i)_k = R_k(c_i) = b_k$$

□

We have now shown that this graph satisfies all necessary constraints to form a valid solution: by Claim 4.4, each individual unit cube contains no two regions with the same color, satisfying Lemma 2.8; additionally, by Claim 4.5, each of these distinct regions is colored such that Theorem 3.2 is satisfied. Thus, a valid solution exists for all n colors and a valid solution has been found for the symmetric case. □

5 Solution Enumeration and Existence

We have rigorously shown that a symmetric set of UC vectors U which to sum B are sufficient for a valid solution to exist. We may now construct a solvable linear algebra system to enumerate these sets for a given n.

5.1 Generalisation to a Linear Algebra System

In order to generalise a linear algebra system to solve for abritrary n, we represent the sum of a given set of UC vectors U by the coefficients of each possible UC vector in the sum. Let these coefficients be represented by a vector $x \in \mathbb{N}^k$ where k is the number of valid unit cube vectors and thus the number of valid unit cube configurations. Notably, k is independent of n for all $n \geq 6$. Additional constraints are also placed on these entries:

$$\sum_{x_i \in x} x_i = n^3 \text{ and } x_i = mn \quad (m \in \mathbb{N})$$

In order to calculate the value of the sum using the coefficient vector x, we define a matrix $A \in \mathbb{N}^{3\times k}$, whose columns are formed by the valid UC configurations for a given n. The coefficient x_k then represents how many of the cube represented by the vector at A_k are present in the solution.

The system's requirements are constrained by the total of the sum, which is the global b-constraint for the cube with side-length n, shown below.

$$\begin{pmatrix} 6n(n-2)^2 \\ 12n(n-2) \\ 8n \end{pmatrix}$$

Finding solutions for a given n then reduces to finding solutions to the linear algebra system defined by $Ax = B$, ensuring that the values of x_i are valid and multiples of n.

$$Ax = B \quad \text{with } x \in \mathbb{N}^k \text{ and } \sum_{x_i \in x} x_i = n^3 \text{ and } x_i = mn \quad (m \in \mathbb{N})$$

Since every entry in a symmetric solution is a multiple of n, it is often more practical to represent x in terms of a new vector, x', where each entry $x'_i = x_i/n$. This simplifies computation for larger values of n, as the system can be divided by n to yield:

$$Ax' = b \quad \text{with } x' \in \mathbb{N}^k \text{ and } \sum_{x'_i \in x'} x'_i = n^2$$

This transforms the problem into a solvable linear algebra system, with solutions that can be enumerated for any n.

5.2 Linear Algebra System for $n \geq 6$

Because of the value of A is independent of n for $n \geq 6$, a consistent method can be developed for solving these cases. The value of A for all $n \geq 6$ is shown below.

$$\begin{bmatrix} 3 & 4 & 6 & 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 1 & 2 & 0 & 3 \\ 1 & 0 & 0 & 1 & 0 & 2 & 0 \end{bmatrix}$$

Therefore, utilizing the linear algebra's simplified form with x', it reduces as such:

$$\begin{bmatrix} 3 & 4 & 6 & 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 1 & 2 & 0 & 3 \\ 1 & 0 & 0 & 1 & 0 & 2 & 0 \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_6 \end{bmatrix} = \begin{pmatrix} 6(n-2)^2 \\ 12(n-2) \\ 8 \end{pmatrix}$$

$$\begin{bmatrix} 3x_0 + 4x_1 + 6x_2 + x_3 + 2x_4 \\ x_1 + x_3 + 2x_4 + 3x_6 \\ x_0 + x_3 + 2x_5 \end{bmatrix} = \begin{pmatrix} 6(n-2)^2 \\ 12(n-2) \\ 8 \end{pmatrix}$$

$$3x_0 + 4x_1 + 6x_2 + x_3 + 2x_4 = 6(n-2)^2$$

$$x_1 + x_3 + 2x_4 + 3x_6 = 12(n-2)$$

$$x_0 + x_3 + 2x_5 = 8$$

This system is currently underdetermined (that is, the number of unknowns exceeds the number of equations.) In order to find a finite number of solutions, it is necessary to constrain the solution space using the previously defined requirements of x':

$$x' \in \mathbb{N}^k \text{ and } \sum_{x'_i \in x'} x'_i = n^2$$

5.3 Proof of Existence for $n \geq 6$

This sections provides a proof for the existence of solutions $n \geq 6$. The reader may independently verify the existence of solutions to the linear algebra system for $2 \leq n \leq 5$. Solutions for these cases do exist, but are not included in the following theorem.

Theorem 5.1. *For any integer $n \geq 6$, there exists a valid solution to the original cube coloring problem.*

Proof. Consider the linear algebra system defined in Section 5.2. We propose a set of values for x_i as follows:

- $x_2 = (n-2)^2$
- $x_5 = 4$
- $x_6 = 4(n-2)$
- $x_i = 0 \quad \forall i \notin \{2, 5, 6\}$

We show these values satisfy all three equations of the linear algebra system:

$$\begin{aligned} 3x_0 + 4x_1 + 6x_2 + x_3 + 2x_4 &= 6((n-2)^2) = 6(n-2)^2 \\ x_1 + x_3 + 2x_4 + 3x_6 &= 3(4(n-2)) = 12(n-2) \\ x_0 + x_3 + 2x_5 &= 2(4n) = 8 \end{aligned}$$

The set of values must also satisfy the summation constraint $\sum_{x_i \in x} x_i = n^2$:

$$\begin{aligned} \sum_{x_i \in x} x_i &= \sum_{n=0}^{6} x_n \\ &= x_2 + x_5 + x_6 \\ &= (n-2)^2 + 4 + 4(n-2) \\ &= (n-2)^2 + 4(n-1) \\ &= n^2 - 4n + 4 + 4n - 4 \\ &= n^2 \end{aligned}$$

This set of values provides a valid, integer solution to the linear algebra system and is symmetric as all values x_i are divisble by n. Thus, by Theorem 4.3, a solution to this system is sufficient to construct a valid solution to the original problem. This shows that there must exist at least one solution for any given $n \geq 6$ to the original cube coloring problem. □

5.4 Computation of Solutions

A SAT solver can now be employed to enumerate all solutions to this system for a given n, and therefore all possible coloring configurations which are solutions to the original problem. A simplified version of the code used in this paper is shown below. This example uses the Python implementation of Google's CP-SAT solver, which is a modern SAT solver implementation that abstracts much of the more complex boilerplate involved in creating a functional linear algebra system. Importantly for this case, it is able to enumerate all solutions of a given system without the requirement for complex solution exclusion criteria to be added manually.

```python
from ortools.sat.python import cp_model

model = cp_model.CpModel()
n = 6 # or any value >= 6

# Define integer variables x0 to x6, restricting their value between 0 and n^2
x = [model.NewIntVar(0, n**2, f'x{i}') for i in range(7)]

# Constraints
model.Add(sum(x) == n**2) # Ensure all values sum to n^2
model.Add(3 * x[0] + 4 * x[1] + 6 * x[2] + x[3] + 2 * x[4] == 6 * (n - 2) * (n - 2))
model.Add(x[1] + x[3] + 2 * x[4] + 3 * x[6] == 12 * (n - 2))
model.Add(x[0] + x[3] + 2 * x[5] == 8)

# Initialise solver and find all solutions (solution_printer is not defined in this example)
solver = cp_model.CpSolver()
solver.SearchForAllSolutions(model, solution_printer)
```

The full version of this program, along with an additional graphing implementation, can be found in the paper's appendices.

6 Results

The SAT solver defined above can be used to generate all solutions to the linear algebra system for any given $n \geq 6$. A plot of these numbers with $6 \leq n \leq 101$ is shown below. The graph shows that there exist many solutions for even small n: there exist 3022 solutions to the linear algebra system with $n = 6$ and climbing to 2,914,300 for $n = 101$.

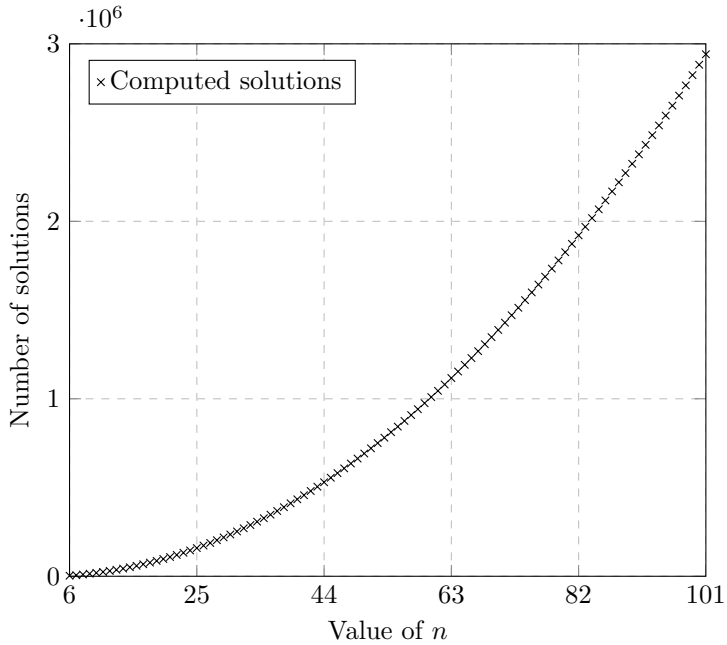

Figure 1: Number of solutions to the linear algebra system for given n

7 Discussion

7.1 Theoretical Considerations

The largest improvement to this project would be a general proof of sufficiency that holds for all cases, as well as symmetric ones. The solution space of symmetric solutions to the linear algebra system is many orders of magnitude smaller than the general solution space without a symmetry constraint: a conservative estimate would be 0.1%. Thus, a general proof without the symmetry constraint would greatly increase the solved scope of the problem. Additionally, a *constructive* proof for coloring of non-symmetric unit cube configurations would expedite the process of generating these colorings manually using a SAT solver and lend additional rigour and completeness to the formal abstraction of the problem, if it were to be solved for the general case.

7.2 Computational Improvements

Other improvements to this project lie in enumerating and displaying real solutions. First, it would be straightforward to allow the SAT solver implementation to generate the value of A for any arbitrary n instead of using a hardcoded value that is only valid for $n \geq 6$. This would allow solutions for any $n \geq 2$ to be enumerated and make the program much more practical. This implementation would constrain solutions via the requirements established in Section 4.

An additional SAT solver implementation would be required in order to practically generate region colorings for a solution to the linear algebra system. This implementation would be a representation of the graph coloring problem introduced in Theorem 4.3, with the color of each node v represented by the variable $x_{v,c}$ where $x_{v,c}$ is true if and only if the node v is colored with the color c.

Finally, in order to fully display real solutions for a given n, an implementation to generate valid arrangements of faces on unit cubes such that they conform to the desired region configuration and coloring would need to be created.

8 Conclusion

We have rigorously abstracted the problem of cube coloring to a solvable linear algebra system and proven the necessity and sufficiency of this abstraction for a specific case of the original problem, where the solution is symmetric. We have shown categorically that solutions for any $n \geq 6$ exist for arbitrarily large n and empirically generated all solutions for $6 \leq n \leq 101$ using an implementation of Google's CP-SAT satisfiability solver. This program, with further modifications outlined in Section 7, could be developed to efficiently enumerate all real solutions to the original problem for a given cube size. Additionally, this paper provides an abstract graph theoretic foundation for the possibility of a general proof for all cases to the problem, which would greatly increase the solution's scope.

9 Appendix

```python
import os
from ortools.sat.python import cp_model

"""
CUBE COLORING SAT-SOLVER
August 2025
Author: Elijah Gallamoy

PACKAGE REQUIREMENTS: ortools (pip install ortools)

DESCRIPTION
This program iterates through all solutions for n between the parameters passed to solve_nums on line 133, which can be modified using the constants defined below.
The program can be configured to generate either symmetric solutions or all solutions (default to symmetric solutions)
The program will currently generate invalid solutions for n < 6.

SAVING
This program assumes the existence of a directory solutions/ in the same scope as where the file is run. It will individually save solutions for each n into separate text files named n.txt.
The first line of the text file is the number of solutions and the following lines are each individual solution as an array representing the solution vector x.

SOLVING FOR LARGER N
Solving for larger values of n will quickly make each solution take an extremely long time to compute if running on regular hardware. As such, it's likely that the solver may crash at some point
during this process and not be able to save the solutions to disc. In order to mitigate this, ensure that the process is not interrupted and do not allow the computer to sleep while the solver is running.
"""

OUT_DIR = 'solutions' # Ensure this directory exists in the same scope that the file is run.

GENERATE_SYMMETRIC = True # If True, generate only symmetric solutions. If False, generate all solutions.
START_VALUE = 6 # Value of n to start from
END_VALUE = 10 # Value of n to end on, inclusively
OUTPUT_EVERY = 1000 # Output progress every OUTPUT_EVERY solutions found.

def solve(n: int) -> tuple[int, list[list[int]]]:
    """Run the ORsolver for a cube with side-length n and returns the solutions.

    Args:
        n (int): the side length of cube to solve for. (range 6 to Inf)

    Returns:
        tuple[int, list[list[int]]]: a tuple containing the number of solutions and a list of the solutions themselves
    """
    print(f'Solving for {n=} length cube...')
    model = cp_model.CpModel()

    multiplier = n if not GENERATE_SYMMETRIC else 1 # Multiply all values by n if generating all solutions to recover the general system from the x' system.

```

```python
    # Define integer variables x1 to x7 (0-indexed here)
    x = [model.NewIntVar(0, n**2 * multiplier, f'x{i}') for i in range(7)]

    # Constraints
    model.Add(sum(x) == n**2 * multiplier)
    model.Add(x[0] + x[3] + 2 * x[5] == 8 * multiplier)
    model.Add(x[1] + x[3] + 2 * x[4] + 3 * x[6] == 12 * (n - 2) * multiplier)
    model.Add(3 * x[0] + 4 * x[1] + 6 * x[2] + x[3] + 2 * x[4] == 6 * (n - 2) * (n - 2) * multiplier)

    # Create solver and define solution printer
    class SolutionPrinter(cp_model.CpSolverSolutionCallback):
        def __init__(self, variables):
            cp_model.CpSolverSolutionCallback.__init__(self)
            self.variables = variables
            self.solution_count = 0
            self.solutions = []

        def on_solution_callback(self):
            solution = [self.Value(v) for v in self.variables]
            self.solutions.append(solution)
            self.solution_count += 1

            # Print out solution every `OUTPUT_EVERY` solutions found
            if self.solution_count % OUTPUT_EVERY == 0:
                print(f"Found {self.solution_count} solutions for {n=}; Sample:", solution)

    solver = cp_model.CpSolver()
    solution_printer = SolutionPrinter(x)

    # Search all solutions
    solver.SearchForAllSolutions(model, solution_printer)

    return (solution_printer.solution_count, solution_printer.solutions)

def write_solution(n: int, count: int, solutions: list[list[int]]) -> None:
    """Save the data of a solution for a given n to disk

    Args:
        n (int): the number that these solutions are for
        count (int): the number of solutions
        solutions (list[list[int]]): the list of solutions themselves
    """

    filename = f'{OUT_DIR}/{n}.txt'

    # Write data to file
    # Format: first line in count, subsequent lines are one solution each.
    with open(filename, 'w') as file:
        file.write(f'{count}\n')
        file.write('\n'.join(str(sol) for sol in solutions))

def get_solved_nums() -> dict[int, int]:
    """
    Return all the numbers that have already been solved from OUT_DIR
    Doesn't currently return all actual solutions from the file, as this would slow down the function significantly.

```

```python
    Returns:
        dict[int, int]: key: value dict of number of solutions with the value of n as the key and the number of solutions as the value
    """

    solved_nums: dict[int, int] = {}
    for filename in os.listdir(OUT_DIR):
        # os.path.splitext splits a path into a (root, ext) tuple.
        # We take the root and append it to our list.
        root, ext = os.path.splitext(filename)
        if ext == '.txt':
            with open(f'{OUT_DIR}/{filename}', 'r') as file:
                count = int(file.readlines()[0].replace('\n', ''))
                solved_nums[int(root)] = count

    return solved_nums

def solve_nums(start: int, end: int) -> None:
    """
    Wrapper around all other functions that solves all numbers between the range start and end, inclusively and saves them to disc,
    skipping over already solved numbers (numbers that already have a file in OUT_DIR).

    Args:
        start (int): the number to start from
        end (int): the number to end on, inclusively
    """
    nums = range(start, end+1)

    # Get rid of any numbers already solved from the list of numbers we want to solve.
    print(f'Getting existing solved numbers...')
    solved_nums = get_solved_nums().keys()

    # Solve for all remaining numbers
    for num in nums:
        if num in solved_nums:
            print(f'Already solved {num}. Skipping...')
            continue

        # Run the solver for the number
        count, solutions = solve(num)

        # Save the solutions to disc
        write_solution(num, count, solutions)

if __name__ == '__main__':
    solve_nums(START_VALUE, END_VALUE)
```

Listing 1: Main Solver Program

```python
"""
Plotter for ORsolver solutions.
Grabs solutions via function from orsolver.py and uses matplotlib to plot them.

PACKAGE REQUIREMENTS: matplotlib (pip install matplotlib)
"""

import matplotlib.pyplot as plt
```

```python
from orsolver import get_solved_nums

# Get and set values
snums = get_solved_nums()
snums = dict(sorted(snums.items(), key=lambda item: item[1])) # Sort values properly by x

y_values = list(snums.values())
x_values = list(snums.keys())

# Plotting
plt.figure(figsize=(10, 6))
plt.plot(x_values, y_values)
plt.title("Possible coloring configurations for cube with size-length n")
plt.xlabel("n")
plt.ylabel("number of solutions")
plt.grid(True)
plt.xticks(x_values)
plt.tight_layout()
plt.show()
```

Listing 2: Graphing Program